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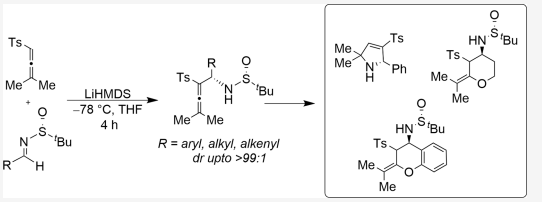
Diastereoselective Addition of Lithiated Allenyl Sulfones to Nonracemic Sulfinimines: Asymmetric Synthesis of Diverse Heterocycles

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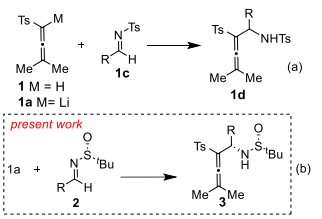
ABSTRACT: The addition of lithiated allenyl sulfones to nonracemic sulfinimines furnished sulfinamides possessing the allenyl sulfonyl moiety with excellent diastereoselectivity. The formed products are transformed into different heterocycles, including the synthesis of pyrroline, pyran, and functionalized benzopyran.



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Nonracemic sulfinimines developed by Davis and Ellman are widely used building blocks for the synthesis of chiral amine-containing compounds.[1] With the consistently high diastereoselectivities realized in the addition of nucleophiles to the sulfinimines and the ease of removal of the sulfinyl group, the sulfinimines have been widely used in the asymmetric synthesis of nitrogen-containing compounds. While a plethora of nucleophilic addition reactions of sulfinimines have been reported in the literature, the addition of simple metalated allenes to sulfinimines has been studied less frequently. The addition of simple allenylzinc bromide or allenyl lithium reagents to sulfinimines always produced the corresponding homopropargyl amines.[2] Although there has been one report of the addition of lithiated sulfonyl allenes to *p*-toluenesulfonyl imines (Scheme 1, eq a), there has been no report on the asymmetric addition to sulfinimines or imines.[3] We reasoned that the lithiation of allenyl sulfone **1** should lead to lithiated allene **1a**[4] stabilized by the directing sulfonyl group that upon addition to nonracemic sulfinimines should form the allene-containing sulfinamides (Scheme 1, eq b) in an asymmetric fashion. Allenes containing the amine/hydroxy tether are proven substrates for the synthesis of several heterocyclic compounds.[5]

Scheme 1. Addition of Lithiated Sulfonylallenes to Nonracemic Sulfinimines



Accordingly, the investigations commenced with the addition of lithiated sulfonyl allene **1a** (prepared by reaction of **1** with LiHMDS) to sulfinimine **2a** obtained from benzaldehyde. It was found that the reaction of sulfone **1** with sulfinimine **2a** in the presence of 1.2 equiv of LiHMDS furnished product **3a** in 83% yield with a >99:1 diastereomeric ratio after column chromatography. It is important to note that the addition of lithiated allene devoid of the methyl groups to **2a** was messy, with no appreciable formation of any characterizable product. Generalization of the reaction with the addition of lithiated allene **1** to several sulfinimines prepared from various aryl aldehydes was undertaken. Thus, addition of **1** to sulfinimine **2b** derived from 2-bromo benzaldehyde gave product **3b** with excellent diastereoselectivity in good yields. The addition of allene **1** to sulfinimine **2c** derived from 2-OMe benzaldehyde afforded a separable mixture of diastereomers **3c** and **3c′** in 69% and 4% yields, respectively, while the addition to sulfinimine **2d** obtained from 2-*tert*-butyl-dimethylsilyloxy benzaldehyde produced a separable mixture of diastereomers **3d** and **3d′** in 25% and 7% yields, respectively. Addition to sulfinimines **2e−g** obtained from 3-OMe benzaldehyde, 4-OMe benzaldehyde, and 4-Cl benzaldehyde furnished allenyl sulfinamides **3e−g**, respectively, in good yields with good to excellent diastereomeric ratios. Addition of **1** to sulfinimine **2h** obtained from 4-nitro benzaldehyde produced product **3h** in 42% yield (dr >99:1). Reaction of **1** with sulfinimine **2i−k** prepared from heteroatom-containing aromatic aldehydes such as 2-thienylcarboxaldehyde, 3-furancarboxaldehyde, and *N*-Boc indole-3-carboxaldehyde furnished products **3i** (dr 94:6), **3j** (dr 94:6), and **3k** (dr 76:24, nonseparable) in 75%, 81%, and 87% yields,

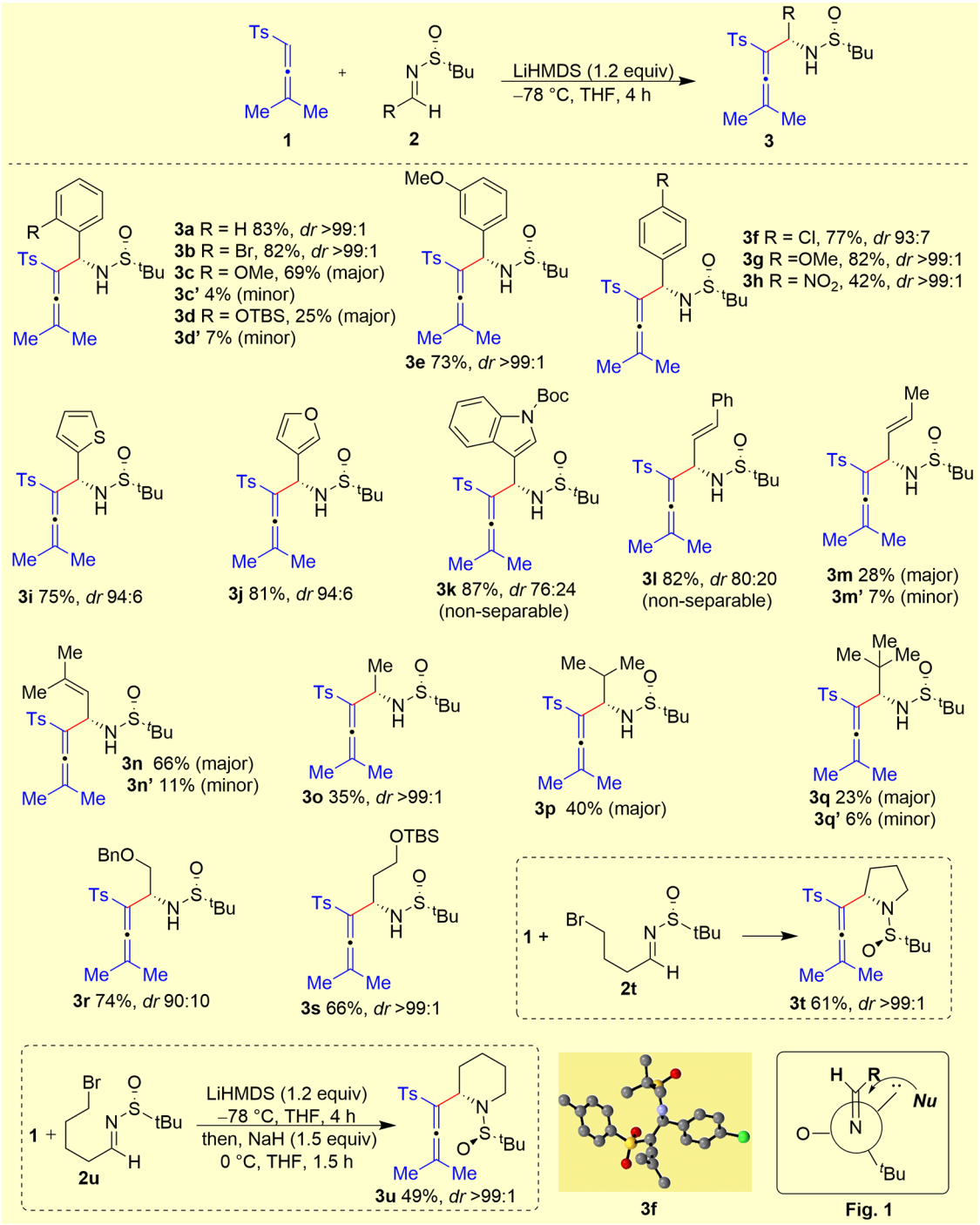
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Scheme 2. Diastereoselective Addition of Lithiated Sulfonylallene 1a to Nonracemic Sulfinimines 2[a,b]



[a]All reactions were carried out with 0.5 mmol of sulfinimine, 0.6 mmol of allene, and 0.6 mmol of LiHMDS in THF (6 mL) as the solvent. [b]The absolute configuration of the newly formed stereogenic center in products **3a–u** was determined to be analogous to that of **3f**, the structure of which was confirmed by X-ray crystal structure analysis.

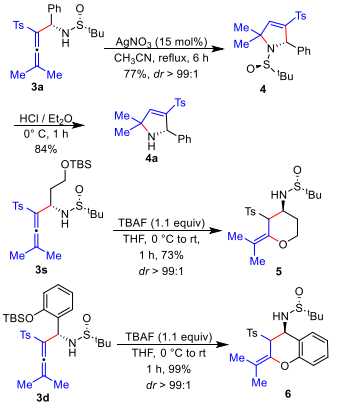
respectively. Addition of allene **1** to sulfinimine **2l** obtained from cinnamaldehyde afforded product **3l** as a nonseparable mixture (dr 80:20) in 82% yield, while reaction with sulfinimine **2m** obtained from crotonaldehyde afforded a separable mixture of diastereomers **3m** and **3m′** in 28% and 7% yields, respectively. However, reaction with sulfinimine **2n** prepared from 3-methylcrotonaldehyde furnished major diastereomer **3n** in 66% yield, while minor diastereomer **3n′** was isolated in 11% yield. The stereochemistry of the formed products was further confirmed by X-ray crystal structure analysis of compound **3f**.[6] After the addition of allene **1** with sulfinimines derived from aromatic aldehydes and unsaturated aldehydes was established, the reaction with sulfinimines prepared from aliphatic aldehydes was investigated. The reaction of **1** with sulfinimine **3o** derived from acetaldehyde furnished the product in 35% yield (dr >99:1), while the reaction with isobutyraldehyde **3p** afforded the major diastereomer in 40% yield after chromatography.[7] Sulfinimine **3q** prepared from pivalaldehyde gave the major

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product in 23% yield, while the minor product was isolated in 6% yield. Addition to sulfinimine **2r** obtained from benzyloxyacetaldehyde furnished allene **3r** as a nonseparable 90:10 mixture of diastereomers. The reaction with sulfinimine **2s** prepared from 3-*tert*-butyldimethylsilyloxy propanaldehyde afforded product **3s** as the major diastereomer in 66% yield. Addition of **1** to sulfinimine **2t** prepared from 4-bromo-butanal directly produced sulfinamido pyrrolidine **3t** in 61% yield. However, the reaction of sulfinimine **2u** prepared from 5-bromo-pentanal afforded a mixture of intermediate sulfinamide and cyclic sulfinamido piperidine **3u**, which was cumbersome to purify. To overcome this difficulty, the product obtained after the addition of **1** to imine **2u** was treated with NaH to afford sulfinamido piperidine **3u** in 49% yield with a >99:1 diastereomeric ratio (Scheme-2). It is evident from the crystal structure analysis that the addition of a nucleophile is taking place from a side opposite that of the sulfinyl oxygen. This can be explained by a nonchelation model proposed earlier by Davis[8a] for the addition of organozinc reagents to sulfinimines and by us in our earlier studies concerning the addition of bulky nucleophiles to sulfinimines (Fig. 1 in Scheme 2).[8b,c]

The utility of the formed sulfinamido allenes in the synthesis of diverse heterocycles was demonstrated by the following transformations. The reaction of **3a** with $AgNO_3$ gave substituted sulfinamido pyrroline **4** in 77% yield.[3] Treatment of **4** with ethereal HCl cleanly removed the sulfinyl group to afford pyrroline **4a** in 84% yield. However, attempted deprotection of the sulfinyl group in allene **3a** was not clean and produced an array of noncharacterizable products. Desilylation of **3s** with TBAF not only deprotected the silylether but also led to an intramolecular Michael addition to afford substituted tetrahydropyran **5** in 73% yield. A similar outcome was observed in the reaction of **3d** with TBAF to afford substituted benzopyran **6** in 99% yield (Scheme 3).[9]

Scheme 3. Transformation of Sulfinamido Allenes to Heterocycles



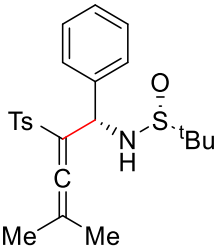
In conclusion, the diastereoselective addition of lithiated sulfonyl allenes to nonracemic sulfinimines was developed. The reaction was found to be general, and sulfinimines prepared from aromatic, aliphatic, and unsaturated aldehydes provided the products with good to excellent diastereomeric ratios and yields. The utility of the formed products was exemplified in the synthesis of diverse heterocycles such as functionalized pyrroline, pyran, and benzopyran.

EXPERIMENTAL SECTION

General Information. Column chromatography was performed on silica gel, Acme grade 100−200 mesh, or 230−400 mesh. Thin layer chromatography (TLC) plates were visualized either with ultraviolet radiation, in an iodine chamber, or with phosphomolybdic acid spray, unless noted otherwise. All of the reagents were purchased from commercial sources and used without additional purification. LiHMDS was purchased as a 1.0 M solution in tetrahydrofuran (THF) and used as received. THF was freshly distilled over Na-benzophenone ketyl. Melting points were uncorrected. Unless stated otherwise, ^{1}H and ^{13}C nuclear magnetic resonance (NMR) spectra were recorded on a 400 MHz machine with $CDCl_3$ as the solvent with TMS or residual $CHCl_3$ as a reference. High-resolution mass spectra (HRMS) were recorded on a Q-TOF micro mass spectrometer using electrospray ionization mode. Allene **1** was prepared according to the procedure described in the literature.[10] All of the sulfinimines were prepared according to the procedure described in the literature.[11]

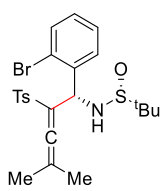
General Procedure for the Addition of Allene to Sulfinimines. The following procedure for the synthesis of **3a** by the addition of allene to sulfinimine **2a** derived from benzaldehyde is representative.

*(S_R)-(S)-1-tert-Butyl-N-(4-methyl-1-phenyl-2-tosylpenta-2,3-dien-1-yl)-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3a**).*



A solution of allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2a** (0.105 g, 0.5 mmol) in dry THF (6 mL) under an argon atmosphere was cooled to −78 °C and stirred for 10 min. LiHMDS (0.6 mL of a 1.0 M solution in THF, 0.6 mmol) was added to the reaction mixture, and it was stirred at the same temperature (−78 °C) for an additional 4 h. After completion of the reaction (TLC), it was quenched by the addition of a saturated aqueous NH_4Cl solution (5 mL), and the mixture was extracted with EtOAc (3 × 10 mL). The organic layer was washed with brine (20 mL) and dried over Na_2SO_4. The crude residue obtained after evaporation of the solvent was purified by silica gel column chromatography using a petroleum ether/EtOAc eluent to afford **3a** in 83% yield (0.180 g) as a pale yellow solid (dr >99:1): R_f = 0.4 (1:1 petroleum ether/EtOAc); mp 78−80 °C; $[\alpha]_D^{24}$ +14.5 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3280, 2926, 1963, 1599, 1072 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.44 (d, *J* = 7.9 Hz, 2H), 7.21−7.19 (m, 5H), 7.12 (d, *J* = 7.9 Hz, 2H), 5.34 (d, *J* = 4.9 Hz, 1H), 3.76 (d, *J* = 4.7 Hz, 1H), 2.34 (s, 3H), 1.75 (s, 3H), 1.69 (s, 3H), 1.15 (s, 9H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 203.7, 144.0, 139.34, 138.2, 129.5, 128.6, 128.2, 127.71, 127.7, 113.71, 109.0, 57.6, 56.3, 22.5, 21.6, 19.6, 19.4; HRMS (ESI) *m*/*z* [M + Na]$^+$ calcd for $C_{23}H_{29}NO_3S_2Na$ 454.1481, found 454.1487.

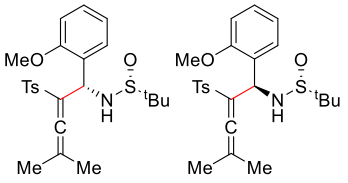
*(S_R)-(S)-N-[1-(2-Bromophenyl)-4-methyl-2-tosylpenta-2,3-dien-1-yl]-1-(tert-butyl)-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3b**).*



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Compound **3b** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2b** (0.144 g, 0.5 mmol) using the general procedure described above in 82% yield (0.210 g) as a white solid (dr >99:1): R_f = 0.4 (1:1 petroleum ether/EtOAc); mp 75−77 °C; $[\alpha]_D^{24}$ −27.2 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3279, 2925, 1963, 1456, 1073 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.48−7.46 (m, 3H), 7.30 (d, *J* = 8.0 Hz, 1H), 7.23 (t, *J* = 7.5 Hz, 1H), 7.12 (d, *J* = 8.1 Hz, 2H), 7.05−7.01 (m, 1H), 5.67 (d, *J* = 7.8 Hz, 1H), 4.43 (d, *J* = 7.8 Hz, 1H), 2.34 (s, 3H), 1.69 (s, 3H), 1.55 (s, 3H), 1.21 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 204.4, 143.9, 137.8, 137.6, 132.9, 130.2, 129.5, 129.3, 127.6, 127.5, 123.1, 111.9, 108.8, 58.4, 56.9, 22.5, 21.6, 19.3, 18.9; HRMS (ESI) *m*/*z* $[M + H]^+$ calcd for $C_{23}H_{29}BrNO_3S_2$ 510.0767, found 510.0772.

*(S_R)-(S)-1-tert-Butyl-N-[1-(2-methoxyphenyl)-4-methyl-2-tosylpenta-2,3-dien-1-yl]-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3c**) and (S_R)-(R)-1-tert-Butyl-N-[1-(2-methoxyphenyl)-4-methyl-2-tosylpenta-2,3-dien-1-yl]-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3c′**).*

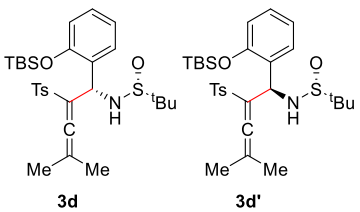


Compounds **3c** and **3c′** were prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2c** (0.120 g, 0.5 mmol) using the general procedure described above as separable diastereomers: major isomer **3c** in 69% yield (0.160 g) as a pale yellow viscous oil and minor isomer **3c′** in 4% yield (0.009 g) as a pale yellow viscous oil.

Data for major isomer **3c**: R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ +37.8 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3277, 2925, 1964, 1604, 1068 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.47 (d, *J* = 8.2 Hz, 2H), 7.16−7.13 (m, 4H), 6.72 (d, *J* = 8.7 Hz, 2H), 5.31 (d, *J* = 4.9 Hz, 1H), 3.75 (s, 3H), 3.67 (d, *J* = 4.8 Hz, 1H), 2.37 (s, 3H), 1.77 (s, 3H), 1.73 (s, 3H), 1.16 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.5, 159.6, 143.9, 138.4, 131.4, 129.4, 129.0, 127.8, 114.0, 113.9, 109.0, 57.0, 56.2, 55.4, 22.6, 21.6, 19.7, 19.5; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{24}H_{31}NO_4S_2Na$ 484.1587, found 484.1591.

Data for minor isomer **3c′**: R_f = 0.5 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −20.0 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3274, 2926, 1970, 1605, 1069 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.67 (d, *J* = 8.2 Hz, 2H), 7.28 (s, 3H), 7.11 (d, *J* = 8.6 Hz, 2H), 6.77 (d, *J* = 8.7 Hz, 2H), 5.17 (d, *J* = 4.3 Hz, 1H), 4.50 (d, *J* = 4.2 Hz, 1H), 3.79 (s, 3H), 2.43 (s, 3H), 1.80 (s, 3H), 1.63 (s, 3H), 1.22 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.7, 159.5, 144.3, 137.8, 130.5, 129.7, 129.3, 128.1, 113.8, 113.6, 109.4, 56.9, 56.1, 55.4, 22.7, 21.8, 19.5, 19.3; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{24}H_{31}NO_4S_2Na$ 484.1587, found 484.1595.

*(S_R)-(S)-1-tert-Butyl-N-(1-{2-[(tert-butyldimethylsilyl)oxy]phenyl}-4-methyl-2-tosylpenta-2,3-dien-1-yl)-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3d**) and (S_R)-(R)-1-tert-Butyl-N-(1-{2-[(tert-butyldimethylsilyl)oxy]phenyl}-4-methyl-2-tosylpenta-2,3-dien-1-yl)-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3d′**).*

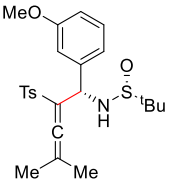


Compounds **3d** and **3d′** were prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2d** (0.170 g, 0.5 mmol) using the general procedure described above as separable diastereomers: major isomer **3d** in 25% yield (0.070 g) as a colorless oil and minor isomer **3d′** in 7% yield (0.020 g) as a colorless oil.

Data for major isomer **3d**: R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −2.0 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 2926, 2860, 1966, 1472, 1078 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.52 (d, *J* = 8.2 Hz, 2H), 7.16−7.13 (m, 3H), 7.09−7.05 (m, 1H), 6.77 (t, *J* = 7.5 Hz, 1H), 6.71 (d, *J* = 8.1 Hz, 1H), 5.86 (d, *J* = 6.1 Hz, 1H), 3.76 (d, *J* = 6.0 Hz, 1H), 2.35 (s, 3H), 1.68 (s, 3H), 1.60 (s, 3H), 1.16 (s, 9H), 1.05 (s, 9H), 0.29 (d, *J* = 4.3 Hz, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.8, 153.0, 143.8, 138.3, 129.3, 129.3, 129.0, 128.98, 127.8 121.1, 118.5, 113.4, 108.8, 56.4, 51.6, 26.1, 22.5, 21.6, 19.6, 19.5, 18.5, −3.9, −3.92; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{29}H_{43}NO_4S_2SiNa$ 584.2295, found 584.2300.

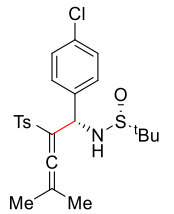
Data for minor isomer **3d′**: R_f = 0.5 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ +16.5 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3264, 2927, 1972, 1470, 1076 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.72 (d, *J* = 8.0 Hz, 2H), 7.26−7.21 (m, 3H), 7.10−7.01 (m, 1H), 6.82 (t, *J* = 7.6 Hz, 1H), 6.70 (d, *J* = 8.0 Hz, 1H), 5.68 (d, *J* = 3.7 Hz, 1H), 4.85 (d, *J* = 3.4 Hz, 1H), 2.40 (s, 3H), 1.68 (s, 3H), 1.58 (s, 3H), 1.21 (s, 9H), 0.87 (s, 9H), 0.15 (d, *J* = 6.3 Hz, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 204.4, 153.4, 144.2, 137.8, 129.8, 129.4, 129.3, 128.8, 128.2, 120.9, 118.6, 112.8, 108.9, 56.1, 50.4, 26.0, 22.7, 21.7, 19.4, 19.3, 18.3, −4.1; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{29}H_{43}NO_4S_2SiNa$ 584.2295, found 584.2300.

*(S_R)-(S)-1-tert-Butyl-N-[1-(3-methoxyphenyl)-4-methyl-2-tosylpenta-2,3-dien-1-yl]-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3e**).*



Compound **3e** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2e** (0.120 g, 0.5 mmol) using the general procedure described above in 73% yield (0.170 g) as a pale yellow oil (dr >99:1): R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ +12.37 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3264, 2923, 1967, 1740, 1070 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.46 (d, *J* = 7.1 Hz, 2H), 7.15−7.09 (m, 3H), 6.84 (d, *J* = 7.3 Hz, 1H), 6.74−6.72 (m, 2H), 5.32 (d, *J* = 5.1 Hz, 1H), 3.76 (d, *J* = 5.1 Hz, 1H), 3.68 (s, 3H), 2.35 (s, 3H), 1.78 (s, 3H), 1.73 (s, 3H), 1.16 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.7, 159.6, 143.9, 140.7, 138.2, 129.6, 129.3, 127.6, 119.9, 113.9, 113.5, 112.9, 108.9, 57.4, 56.2, 55.1, 22.4, 21.5, 19.5, 19.3; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{24}H_{31}NO_4S_2Na$ 484.1587, found 484.1592.

*(S_R)-(S)-1-tert-Butyl-N-[1-(4-chlorophenyl)-4-methyl-2-tosylpenta-2,3-dien-1-yl]-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3f**).*

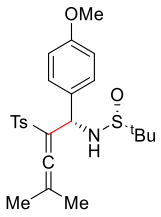


Compound **3f** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2f** (0.144 g, 0.5 mmol) using the general procedure described above in 77% yield (0.180 g) as a pale yellow solid (dr 93:7): R_f = 0.4 (1:1 petroleum ether/EtOAc); mp 135−137 °C; $[\alpha]_D^{24}$ +38.5 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3275, 2923, 1970, 1598, 1081 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.47 (d, *J* = 8.2 Hz, 2H), 7.18−7.17 (m, 6H), 5.33 (d, *J* = 5.8 Hz, 1H), 3.84 (d, *J* = 5.8 Hz, 1H), 2.39 (s, 3H), 1.78 (s,

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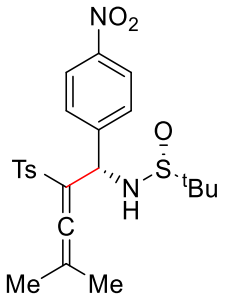
3H), 1.73 (s, 3H), 1.18 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.5, 144.3, 138.2, 138.0, 134.2, 129.6, 129.1, 128.8, 127.8, 113.5, 109.3, 57.3, 56.5, 22.6, 21.7, 19.7, 19.5; HRMS (ESI) m/z $[M + Na]^+$ calcd for $C_{23}H_{28}ClNO_3S_2Na$ 488.1091, found 488.1095.

*(S_R)-(S)-1-tert-Butyl-N-[1-(4-methoxyphenyl)-4-methyl-2-tosylpenta-2,3-dien-1-yl]-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3g**).*



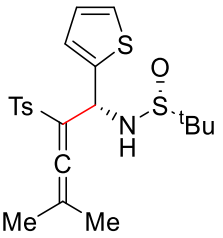
Compound **3g** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2g** (0.120 g, 0.5 mmol) using the general procedure described above in 82% yield (0.190 g) as a pale yellow oil (dr >99:1): R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ +35.3 (c 1.0, $CHCl_3$); IR (neat) ν_{max} 3269, 2925, 1977, 1739, 1070 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.46 (d, J = 8.1 Hz, 2H), 7.14 (d, J = 7.7 Hz, 4H), 6.71 (d, J = 8.5 Hz, 2H), 5.30 (d, J = 4.8 Hz, 1H), 3.74 (s, 3H), 3.67 (d, J = 4.7 Hz, 1H), 2.35 (s, 3H), 1.76 (s, 3H), 1.72 (s, 3H), 1.15 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.4, 159.4, 143.7, 138.2, 131.2, 129.3, 128.7, 127.6, 113.9, 113.7, 108.9, 56.7, 56.1, 55.2, 22.4, 21.5, 19.5, 19.4; HRMS (ESI) m/z $[M + Na]^+$ calcd for $C_{24}H_{31}NO_4S_2Na$ 484.1587, found 484.1591.

*(S_R)-(S)-1-tert-Butyl-N-[4-methyl-1-(4-nitrophenyl)-2-tosylpenta-2,3-dien-1-yl]-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3h**).*



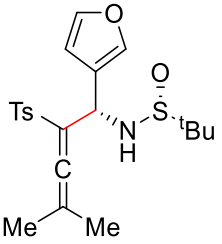
Compound **3h** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2h** (0.127 g, 0.5 mmol) using the general procedure described above in 42% yield (0.100 g) as a pale yellow viscous oil (dr >99:1): R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ +27.5 (c 1.0, $CHCl_3$); IR (neat) ν_{max} 3269, 2924, 1967, 1732, 1074 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 8.06 (d, J = 8.7 Hz, 2H), 7.50 (d, J = 8.3 Hz, 2H), 7.45 (d, J = 8.7 Hz, 2H), 7.18 (d, J = 8.1 Hz, 2H), 5.45 (d, J = 6.8 Hz, 1H), 4.11 (d, J = 6.8 Hz, 1H), 2.38 (s, 3H), 1.77 (s, 3H), 1.69 (s, 3H), 1.19 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.5, 147.6, 146.8, 144.7, 137.8, 129.7, 128.6, 127.8, 123.8, 113.0, 110.0, 57.8, 56.9, 22.5, 21.7, 19.6, 19.4; HRMS (ESI) m/z $[M + H]^+$ calcd for $C_{23}H_{28}N_2O_5S_2Na$ 499.1332, found 499.1337.

*(S_R)-(S)-1-tert-Butyl-N-[4-methyl-1-(thiophen-2-yl)-2-tosylpenta-2,3-dien-1-yl]-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3i**).*



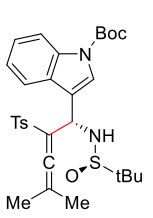
Compound **3i** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2i** (0.108 g, 0.5 mmol) using the general procedure described above in 75% yield (0.165 g) as a pale yellow solid (dr 94:6): R_f = 0.4 (1:1 petroleum ether/EtOAc); mp 76−78 °C; $[\alpha]_D^{24}$ +15.1 (c 1.0, $CHCl_3$); IR (neat) ν_{max} 3274, 2923, 1964, 1737, 1073 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.56−7.54 (m, 2H), 7.18 (d, J = 7.4 Hz, 2H), 7.13−7.12 (m, 1H), 6.97 (s, 1H), 6.82−6.80 (m, 1H), 5.59 (d, J = 5.6 Hz, 1H), 3.85 (d, J = 5.6 Hz, 1H), 2.35 (s, 3H), 1.73 (d, J = 9.0 Hz, 6H), 1.13 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 202.9, 144.0, 143.4, 138.0, 129.4, 127.7, 126.8, 126.7, 126.1, 113.8, 110.0, 56.4, 53.4, 29.7, 22.4, 21.6, 19.4, 19; HRMS (ESI) m/z $[M + Na]^+$ calcd for $C_{21}H_{27}NO_3S_3Na$ 460.1045, found 460.1051.

*(S_R)-(S)-1-tert-Butyl-N-[1-(furan-3-yl)-4-methyl-2-tosylpenta-2,3-dien-1-yl]-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3j**).*



Compound **3j** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2j** (0.100 g, 0.5 mmol) using the general procedure described above in 81% yield (0.170 g) as a colorless oil (dr 94:6): R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −5.8 (c 1.0, $CHCl_3$); IR (neat) ν_{max} 3280, 2923, 1971, 1748, 1070 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.66 (d, J = 7.3 Hz, 2H), 7.36 (s, 1H), 7.28−7.26 (m, 3H), 6.23 (s, 1H), 5.29 (d, J = 6.6 Hz, 1H), 3.78 (d, J = 6.4 Hz, 1H), 2.42 (s, 3H), 1.75 (s, 3H), 1.72 (s, 6H), 1.17 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.0, 144.3, 143.4, 140.8, 138.0, 129.6, 127.7, 125.2, 113.3, 109.5, 109.4, 56.3, 50.8, 22.5, 21.6, 19.4, 19.41; HRMS (ESI) m/z $[M + Na]^+$ calcd for $C_{21}H_{27}NO_4S_2Na$ 444.1274, found 444.1276.

*(S_R)-tert-Butyl (S)-3-(1-{[tert-Butyl(λ^1-oxidaneyl)-λ^3-sulfaneyl]-amino}-4-methyl-2-tosylpenta-2,3-dien-1-yl)-1H-indole-1-carboxylate (**3k**).*

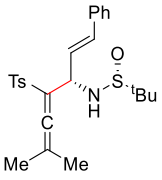


Compound **3k** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2k** (0.174 g, 0.5 mmol) using the general procedure described above in 87% yield (0.250 g) as a colorless viscous oil as a nonseparable mixture of diastereomers (dr

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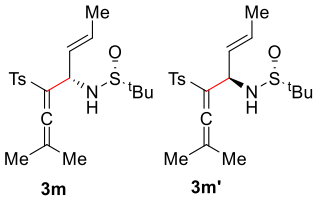
76:24): R_f = 0.5 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ +14.4 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 2969, 1963, 1732, 1454, 1076 cm^{-1}. Data for the major diastereomer: ^1H NMR (400 MHz, $CDCl_3$) δ 7.98 (d, *J* = 7.9 Hz, 1H), 7.51 (s, 1H), 7.39 (dd, *J* = 15.1, 7.9 Hz, 3H), 7.22 (t, *J* = 7.7 Hz, 1H), 7.09 (t, *J* = 7.6 Hz, 1H), 6.94 (d, *J* = 7.9 Hz, 2H), 5.62 (d, *J* = 4.6 Hz, 1H), 3.99 (d, *J* = 4.5 Hz, 1H), 2.27 (s, 3H), 1.87 (s, 3H), 1.81 (s, 3H), 1.67 (s, 9H), 1.19 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.5, 149.2, 143.6, 137.6, 128.8, 127.2, 125.4, 124.5, 122.4, 119.3, 115.2, 111.5, 108.9, 83.9, 56.2, 50.6, 28.1, 22.5, 21.4, 19.6, 19.5; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{30}H_{38}N_2O_5S_2Na$ 593.2114, found 593.2122.

*(S_R)-(S,E)-1-tert-Butyl-N-(6-methyl-1-phenyl-4-tosylhepta-1,4,5-trien-3-yl)-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3l**).*



Compound **3l** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2l** (0.118 g, 0.5 mmol) using the general procedure described above in 82% yield (0.187 g) as a white solid (nonseparable mixture of diastereomers, dr 80:20): R_f = 0.4 (1:1 petroleum ether/EtOAc); mp 115−117 °C; $[\alpha]_D^{24}$ +10.1 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3460, 3272, 2924, 1967, 1458, 1146 cm^{-1}; representative peaks for the major diastereomer; ^1H NMR (400 MHz, $CDCl_3$) δ 7.72 (d, *J* = 8.2 Hz, 2H), 7.30−7.23 (m, 7H), 6.57 (d, *J* = 15.8 Hz, 1H), 6.08 (dd, *J* = 15.8, 7.2 Hz, 1H), 4.91 (t, *J* = 7.0 Hz, 1H), 3.74 (d, *J* = 6.6 Hz, 1H), 2.37 (s, 3H), 1.78 (s, 3H), 1.76 (s, 3H), 1.19 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 202.7, 144.3, 138.5, 136.1, 133.1, 129.6, 128.5, 128.1, 127.7, 126.8, 112.8, 109.4, 56.8, 56.3, 22.7, 22.5, 21.6, 19.6, 19.56; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{25}H_{31}NO_3S_2Na$ 480.1638, found 480.1642.

*(S_R)-(S,E)-1-tert-Butyl-N-(7-methyl-5-tosylocta-2,5,6-trien-4-yl)-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3m**) and (S_R)-(R,E)-1-tert-Butyl-N-(7-methyl-5-tosylocta-2,5,6-trien-4-yl)-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3m′**).*

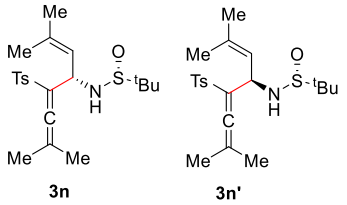


Compounds **3m** and **3m′** were prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2m** (0.087 g, 0.5 mmol) using the general procedure described above as separable diastereomers: major isomer **3m** in 28% yield (0.055 g) as a colorless oil and minor isomer **3m′** in 7% yield (0.014 g) as a colorless oil.

Data for major isomer **3m**: R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −10.4 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3273, 2926, 1970, 1454, 1071 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.71 (d, *J* = 7.2 Hz, 2H), 7.29 (d, *J* = 7.5 Hz, 2H), 5.74−5.67 (m, 1H), 5.42 (dd, *J* = 15.2, 7.3 Hz, 1H), 4.67 (t, *J* = 6.9 Hz, 1H), 3.54 (d, *J* = 6.4 Hz, 1H), 2.41 (s, 3H), 1.71 (s, 6H), 1.61 (d, *J* = 6.4 Hz, 3H), 1.14 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 202.6, 144.2, 138.3, 129.8, 129.7, 129.6, 128.1, 113.1, 109.1, 56.7, 56.2, 22.5, 21.7, 19.6, 17.8; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{20}H_{29}NO_3S_2Na$ 418.1481, found 418.1486.

Data for minor isomer **3m′**: R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −47.8 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3277, 2929, 1965, 1454, 1145 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.77 (d, *J* = 8.2 Hz, 2H), 7.31 (d, *J* = 8.1 Hz, 2H), 5.72−5.63 (dm, 1H), 5.38−5.33 (m, 1H), 4.55 (t, *J* = 6.2 Hz, 1H), 4.12 (d, *J* = 5.3 Hz, 1H), 2.42 (s, 3H), 1.77 (s, 3H), 1.72 (s, 3H), 1.65 (d, *J* = 5.9 Hz, 3H), 1.19 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 202.6, 144.3, 137.9, 130.0, 129.7, 128.5, 128.3, 112.7, 109.2, 56.1, 55.9, 22.7, 21.8, 19.6, 19.4, 17.8; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{20}H_{29}NO_3S_2Na$ 418.1481, found 418.1488.

*(S_R)-(S)-1-tert-Butyl-N-(2,7-dimethyl-5-tosylocta-2,5,6-trien-4-yl)-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3n**) and (S_R)-(R)-1-tert-Butyl-N-(2,7-dimethyl-5-tosylocta-2,5,6-trien-4-yl)-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3n′**).*

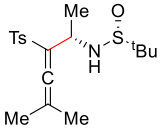


Compounds **3n**and **3n′** were prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2n** (0.094 g, 0.5 mmol) using the general procedure described above as separable diastereomers: major isomer **3n** in 66% yield (0.135 g) as a colorless viscous oil and minor isomer **3n′** in 11% yield (0.023 g) as a colorless viscous oil.

Data for major isomer **3n**: R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −18.6 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3274, 2923, 1963, 1733, 1065 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.71 (d, *J* = 8.2 Hz, 2H), 7.29 (d, *J* = 8.1 Hz, 2H), 5.14 (d, *J* = 8.9 Hz, 1H), 4.92 (dd, *J* = 8.8, 5.8 Hz, 1H), 3.60 (d, *J* = 5.6 Hz, 1H), 2.42 (s, 3H), 1.71 (s, 7H), 1.65−1.63 (m, 6H), 1.15 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 202.4, 144.1, 138.5, 137.1, 129.5, 128.1, 123.5, 113.5, 108.9, 56.1, 52.5, 29.8, 25.8, 22.6, 21.7, 19.6, 19.55, 18.5; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{21}H_{31}NO_3S_2Na$ 432.1638, found 432.1646.

Data for minor isomer **3n′**: R_f = 0.5 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −12.3 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3282, 2926, 1969, 1733, 1074 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.76 (d, *J* = 8.3 Hz, 2H), 7.31 (d, *J* = 8.2 Hz, 2H), 5.03 (d, *J* = 8.8 Hz, 1H), 4.78 (dd, *J* = 8.8, 4.3 Hz, 1H), 4.21 (d, *J* = 4.2 Hz, 1H), 2.42 (s, 3H), 1.79 (s, 3H), 1.72 (s, 3H), 1.65 (s, 3H), 1.51 (1, 3H), 1.19 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 202.6, 144.3, 138.0, 137.6, 129.7, 128.1, 122.3, 112.9, 108.8, 55.7, 51.9, 29.8, 25.8, 22.7, 21.7, 19.6, 19.4, 18.3; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{21}H_{31}NO_3S_2Na$ 432.1638, found 432.1645.

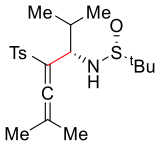
*(S_R)-(S)-1-tert-Butyl-N-(5-methyl-3-tosylhexa-3,4-dien-2-yl)-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3o**).*



Compound **3o** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2o** (0.074 g, 0.5 mmol) using the general procedure described above in 35% yield (0.065 g) as a pale yellow solid (dr >99:1): R_f = 0.4 (1:1 petroleum ether/EtOAc); mp 114−116 °C; $[\alpha]_D^{24}$ −97.1 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3277, 2930, 1964, 1720, 1082 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.77 (d, *J* = 8.1 Hz, 2H), 7.31 (d, *J* = 8.1 Hz, 2H), 4.34 (p, *J* = 6.7 Hz, 1H), 3.61 (d, *J* = 6.5 Hz, 1H), 2.42 (s, 3H), 1.68 (s, 3H), 1.66 (s, 3H), 1.45 (d, *J* = 6.8 Hz, 3H), 1.16 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 201.3, 144.3, 137.7, 129.7, 128.2, 114.8, 109.5, 56.1, 50.4, 22.6, 22.5, 21.7, 19.6, 19.3; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{18}H_{27}NO_3S_2Na$ 392.1325, found 392.1333.

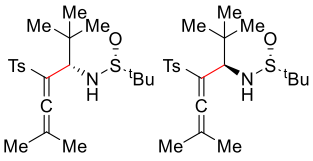
https://doi.org/10.1021/acs.joc.4c01982
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*(S_R)-(S)-1-tert-Butyl-N-(2,6-dimethyl-4-tosylhepta-4,5-dien-3-yl)-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3p**).*



Compound **3p** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2p** (0.088 g, 0.5 mmol) using the general procedure described above in 40% yield (0.078 g) as a colorless oil: R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −18.9 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3297, 2923, 1967, 1458, 1067 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.72 (d, *J* = 8.3 Hz, 2H), 7.28 (d, *J* = 8.1 Hz, 2H), 3.86 (t, *J* = 7.0 Hz, 1H), 3.62 (d, *J* = 6.8 Hz, 1H), 2.39 (s, 3H), 2.14−2.09 (m, 1H), 1.68 (s, 3H), 1.63 (s, 3H), 1.10 (s, 9H), 0.93−0.91 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 202.4, 144.2, 137.8, 129.6, 128.1, 112.8, 108.3, 60.8, 56.2, 32.4, 22.5, 21.7, 19.8, 19.5, 19.1, 18.3; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{20}H_{31}NO_3S_2Na$ 420.1638, found 420.1646.

*(S_R)-(S)-1-tert-Butyl-1-(λ^1-oxidaneyl)-N-(2,2,6-trimethyl-4-tosylhepta-4,5-dien-3-yl)-λ^3-sulfanamine (**3q**) and (S_R)-(R)-1-tert-Butyl-1-(λ^1-oxidaneyl)-N-(2,2,6-trimethyl-4-tosylhepta-4,5-dien-3-yl)-λ^3-sulfanamine (**3q′**).*



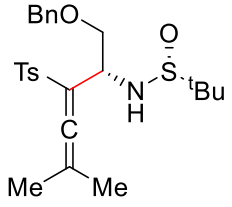
3q **3q'**

Compounds **3q** and **3q′** were prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2q** (0.095 g, 0.5 mmol) using the general procedure described above as separable diastereomers: major isomer **3q** in 23% yield (0.048 g) as a pale yellow viscous oil and minor isomer **3q′** in 6% yield (0.013 g) as a pale yellow viscous oil.

Data for major isomer **3q**: R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ +1.4 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3310, 2924, 1961, 1723, 1070 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.74 (d, *J* = 8.2 Hz, 2H), 7.29 (d, *J* = 8.0 Hz, 2H), 4.06 (d, *J* = 7.0 Hz, 1H), 3.45 (d, *J* = 7.0 Hz, 1H), 2.41 (s, 3H), 1.72 (s, 3H), 1.55 (s, 3H), 1.06 (s, 9H), 1.00 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 204.1, 144.2, 138.3, 129.6, 128.4, 113.7, 108.0, 62.1, 56.1, 36.3, 26.7, 22.6, 21.7, 19.7, 19.0; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{21}H_{33}NO_3S_2Na$ 434.1794, found 434.1801.

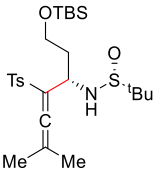
Data for minor isomer **3q′**: R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −44.2 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3288, 2926, 1966, 1727, 1074 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.88 (d, *J* = 8.3 Hz, 2H), 7.29 (d, *J* = 8.1 Hz, 2H), 4.01 (d, *J* = 11.0 Hz, 1H), 3.61 (d, *J* = 11.1 Hz, 1H), 2.39 (s, 3H), 1.58 (s, 3H), 1.42 (s, 3H), 1.20 (s, 9H), 1.03 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.3, 143.9, 138.2, 129.4, 128.6, 113.8, 109.1, 64.1, 56.6, 36.6, 26.8, 22.7, 21.7, 19.3, 18.4; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{21}H_{33}NO_3S_2Na$ 434.1794, found 434.1803.

*(S_R)-(S)-N-[1-(Benzyloxy)-5-methyl-3-tosylhexa-3,4-dien-2-yl]-1-(tert-butyl)-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3r**).*



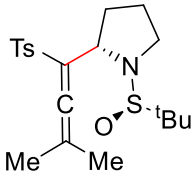
Compound **3r** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2r** (0.127 g, 0.5 mmol) using the general procedure described above in 74% yield (0.175 g) as a colorless oil (dr 90:10): R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −46.4 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3283, 2923, 1966, 1738, 1075 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.76 (d, *J* = 8.1 Hz, 2H), 7.36−7.28 (m, 7H), 4.55−4.52 (m, 1H), 4.46−4.41 (m, 2H), 3.95 (d, *J* = 6.1 Hz, 1H), 3.83 (dd, *J* = 9.8, 4.9 Hz, 1H), 3.64 (dd, *J* = 9.7, 6.5 Hz, 1H), 2.44 (s, 3H), 1.71 (s, 3H), 1.70 (s, 3H), 1.16 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.4, 144.2, 137.7, 137.6, 129.6, 128.3, 127.8, 127.6, 127.6, 110.9, 108.8, 72.8, 72.3, 55.9, 53.3, 22.4, 21.6, 19.4, 19.3; HRMS (ESI) *m*/*z* $[M + H]^+$ calcd for $C_{25}H_{34}NO_4S_2$ 476.1924, found 476.1926.

*(S_R)-(S)-1-tert-Butyl-N-{1-[(tert-butyldimethylsilyl)oxy]-6-methyl-4-tosylhepta-4,5-dien-3-yl}-1-(λ^1-oxidaneyl)-λ^3-sulfanamine (**3s**).*



Compound **3s** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2s** (0.146 g, 0.5 mmol) using the general procedure described above in 66% yield (0.150 g) as a colorless oil (dr >99:1): R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −51.5 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3284, 2932, 1967, 1600, 1081 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.74 (d, *J* = 8.0 Hz, 2H), 7.28 (d, *J* = 8.0 Hz, 2H), 4.36−4.30 (m, 2H), 3.73−3.78 (m, 1H), 3.63−3.58 (m, 1H), 2.39 (s, 3H), 2.01 (bs, 2H), 1.68 (s, 3H), 1.64 (s, 3H), 1.11 (s, 9H), 0.85 (s, 9H), 0.16 to −0.16 (m, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 202.1, 144.2, 137.7, 129.6, 128.1, 113.4, 108.8, 60.4, 55.8, 52.4, 38.0, 26.0, 22.6, 21.7, 19.6, 19.2, 18.4, −5.26, −5.28; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{25}H_{43}NO_4S_2SiNa$ 536.2295, found 536.2299.

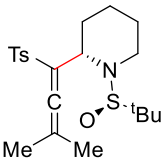
*(S_R)-(S)-1-[tert-Butyl(λ^1-oxidaneyl)-λ^3-sulfaneyl]-2-(3-methyl-1-tosylbuta-1,2-dien-1-yl)pyrrolidine (**3t**).*



Compound **3t** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2t** (0.127 g, 0.5 mmol) using the general procedure described above in 61% yield (0.145 g) as a pale yellow oil (dr >99:1): R_f = 0.4 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −5.3 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 2922, 2862, 1965, 1455, 1304, 1070, 773 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.72 (d, *J* = 8.0 Hz, 2H), 7.31 (d, *J* = 8.0 Hz, 2H), 4.51−4.49 (m, 1H), 3.67−3.61 (m, 1H), 2.73−2.67 (m, 1H), 2.43 (s, 3H), 2.05−2.00 (m, 1H), 1.85−1.78 (m, 3H), 1.74 (s, 3H), 1.68 (s, 3H), 1.05 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 202.2, 144.2, 138.1, 129.6, 127.9, 114.7, 108.1, 64.5, 57.2, 41.0, 33.2, 24.9, 23.5, 21.6, 19.7, 19.6; HRMS (ESI) *m*/*z* $[M + Na]^+$ calcd for $C_{20}H_{29}NO_3S_2Na$ 418.1481, found 418.1490.

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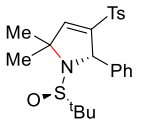
*(S_R)-1-[tert-Butyl(λ^1-oxidaneyl)-λ^3-sulfaneyl]-2-(3-methyl-1-tosylbuta-1,2-dien-1-yl)piperidine (**3u**).*



Compound **3u** was prepared from allene **1** (0.133 g, 0.6 mmol) and sulfinimine **2u** (0.134 g, 0.5 mmol) using the general procedure, and then to a crude reaction mixture in dry THF (5 mL) under an argon atmosphere at 0 °C was added NaH (0.030 g, 0.75 mmol). The reaction mixture was stirred for 1.5 h at the same temperature. After the completion of the reaction, it was quenched with ice-cold water (5 mL), and the mixture was extracted with EtOAc. The organic layer was washed with brine (5 mL) and dried over Na_2SO_4. Evaporation of the solvent followed by silica gel column chromatography of the resulting crude residue with a petroleum ether/EtOAc eluent afforded the compound in 49% yield (0.100 g) as a colorless oil (dr >99:1): R_f = 0.6 (7:3 petroleum ether/EtOAc); $[\alpha]_D^{24}$ +23.8 (c 1.0, $CHCl_3$); IR (neat) ν_{max} 3459, 2929, 1966, 1451, 1078 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.73 (d, J = 8.0 Hz, 2H), 7.31 (d, J = 8.0 Hz, 2H), 4.42 (bs, 1H), 3.36−3.33 (m, 1H), 3.01−2.94 (m, 1H), 2.43 (s, 3H), 1.98−1.88 (m, 2H), 1.77 (s, 3H), 1.64 (s, 2H), 1.62 (s, 3H), 1.50−1.49 (m, 2H), 1.01 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 203.3, 144.2, 137.8, 129.6, 128.0, 112.5, 107.4, 58.5, 56.6, 40.0, 30.3, 26.2, 22.9, 21.7, 20.0, 19.3, 19.1; HRMS (ESI) m/z $[M + Na]^+$ calcd for $C_{21}H_{31}NO_3S_2Na$ 432.1638, found 432.1642.

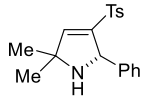
Procedure for the 2.0 mmol Scale Experiment (3a). A solution of allene **1** (0.532 g, 2.4 mmol) and sulfinimine **2a** (0.420 g, 2.0 mmol) in dry THF (20 mL) under an argon atmosphere was cooled to −78 °C and stirred for 10 min. LiHMDS (2.4 mL of a 1.0 M solution in THF, 2.4 mmol) was added to the reaction mixture, and it was stirred at the same temperature (−78 °C) for an additional 4 h. After completion of the reaction (TLC), it was quenched by the addition of a saturated aqueous NH_4Cl solution (10 mL), and the mixture was extracted with EtOAc (3 × 10 mL). The organic layer was washed with brine (20 mL) and dried over Na_2SO_4. The crude residue obtained after evaporation of the solvent was purified by silica gel column chromatography using a petroleum ether/EtOAc eluent to afford **3a** in 77% yield (0.668 g) as a pale yellow solid (dr >99:1).

Product Functionalization. *(S_R)-(S)-1-[tert-Butyl(λ^1-oxidaneyl)-λ^3-sulfaneyl]-2,2-dimethyl-5-phenyl-4-tosyl-2,5-dihydro-1H-pyrrole (**4**).*



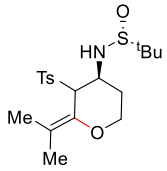
To a stirred solution of allenyl sulfinamide **3a** (0.040 g, 0.093 mmol) in acetonitrile (3 mL) under an argon atmosphere was added $AgNO_3$ (3.0 mg, 0.02 mmol) at room temperature. The reaction mixture was stirred at 85 °C for 6 h. After the completion of the reaction (TLC), the reaction mixture was cooled to room temperature. The solvent was evaporated under reduced pressure, and the crude residue thus obtained was purified by silica gel column chromatography with a petroleum ether/EtOAc eluent to afford **4** in 77% yield (0.032 g) as a white solid (dr >99:1): R_f = 0.5 (1:1 petroleum ether/EtOAc); mp 183−185 °C; $[\alpha]_D^{24}$ +135.5 (c 1.0, $CHCl_3$); IR (neat) ν_{max} 2926, 1607, 1460, 1310, 1150, 953 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.19 (d, J = 8.0 Hz, 2H), 7.13−7.04 (m, 5H), 6.99 (d, J = 8.0 Hz, 2H), 6.73 (s, 1H), 5.91−5.90 (m, 1H), 2.32 (s, 3H), 1.69 (s, 3H), 1.51 (s, 3H), 0.72 (s, 9H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 145.6, 143.9, 142.9, 139.2, 136.8, 129.8, 129.4, 128.3, 128.2, 127.7, 71.8, 61.3, 56.9, 30.5, 26.8, 23.7, 21.6; HRMS (ESI) m/z $[M + Na]^+$ calcd for $C_{23}H_{29}NO_3S_2Na$ 454.1481, found 454.1488.

*(S)-2,2-Dimethyl-5-phenyl-4-tosyl-2,5-dihydro-1H-pyrrole (**4a**).*



To a stirred solution of **4** (0.025 g, 0.058 mmol) in Et_2O (2 mL) at 0 °C was added a saturated solution of ethereal HCl (w/v) (1.0 mL). The reaction mixture was stirred for 1 h at the same temperature. After completion of the reaction (TLC), the solvent was evaporated off, the resultant amine hydrochloride salt was dissolved in DCM (5 mL), and a saturated $NaHCO_3$ solution (2.0 mL) was added. The reaction mixture was stirred for 1 h at room temperature. The reaction mixture was diluted with water (5 mL) and extracted with DCM. The organic layers were washed with brine and dried over Na_2SO_4. Evaporation of the solvent followed by silica gel column chromatography of the resulting crude residue with a petroleum ether/EtOAc el uent afforded **4a** in 84% yield (0.016 g) as a white solid: R_f = 0.4 (1:1 petroleum ether/EtOAc); mp 180−182 °C; $[\alpha]_D^{24}$ +44.3 (c 1.0, $CHCl_3$); IR (neat) ν_{max} 3348, 2923, 2856, 1307, 1146, 1092, 771 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.24 (m, J = 7.23−7.25 Hz, 2H), 7.14 (m, J = 7.12−7.16 Hz, 1H), 7.07 (t, J = 7.3 Hz, 2H), 7.00 (m, J = 6.99−7.01 Hz, 4H), 6.94 (s, 1H), 5.22 (s, 1H), 2.34 (s, 3H), 1.96 (s, 1H), 1.43 (s, 3H), 1.35 (s, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 150.5, 143.9, 143.9, 139.9, 136.9, 129.4, 128.5, 128.2, 127.9, 127.89, 68.0, 65.4, 29.2, 28.8, 21.6; HRMS (ESI) m/z $[M + H]^+$ calcd for $C_{19}H_{22}NO_2S$ 328.1366, found 328.1371.

*(S_R)-(4S)-N-[tert-Butyl(λ^1-oxidaneyl)-λ^3-sulfaneyl]-2-(propan-2-ylidene)-3-tosyltetrahydro-2H-pyran-4-amine (**5**).*

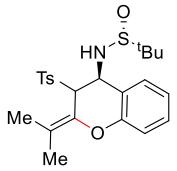


To a solution of allenyl sulfinamide silyl ether **3s** (0.060 g, 0.12 mmol, 1 equiv) in dry THF (4 mL) cooled to 0 °C was added TBAF (0.132 mL of a 1 M solution in THF, 0.132 mmol). The resulting solution was allowed to warm to room temperature (∼1 h) and diluted with CH_2Cl_2 (5 mL). The reaction mixture was poured into water (5 mL). After extraction with CH_2Cl_2 (2 × 5 mL), the organic layer was extracted with brine and dried over Na_2SO_4. Evaporation of the solvent followed by silica gel column chromatography of the resulting residue with a petroleum ether/EtOAc eluent afforded **5** in 73% yield (0.035 g) as a colorless oil (dr >99:1): R_f = 0.6 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −144.8 (c 1.0, $CHCl_3$); IR (neat) ν_{max} 3203, 2926, 1737, 1465, 1057 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$) δ 7.73 (d, J = 8.0 Hz, 2H), 7.31 (d, J = 8.0 Hz, 2H), 4.44 (s, 1H), 4.18 (s, 1H), 4.09−4.06 (m, 1H), 3.82−3.77 (m, 1H), 3.51 (d, J = 5.8 Hz, 1H), 2.70−2.62 (m, 1H), 2.43 (s, 3H), 1.81−1.77 (m, 1H), 1.59 (s, 3H), 1.17 (s, 9H), 1.00 (s, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 145.1, 135.3, 135.0, 129.7, 129.2, 124.1, 65.5, 63.9, 56.3, 48.6, 28.8, 22.6, 21.8, 17.9, 17.4; HRMS (ESI)

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m/*z* [M + Na]$^+$ calcd for $C_{19}H_{29}NO_4S_2Na$ 422.1430, found 422.1436.

*(S_R)-(4S)-N-[tert-Butyl(λ^1-oxidaneyl)-λ^3-sulfaneyl]-2-(propan-2-ylidene)-3-tosylchroman-4-amine (**6**).*



To a solution of allenyl sulfinamide silyl ether **3d** (0.038 g, 0.068 mmol, 1 equiv) in dry THF (3 mL) cooled to 0 °C was added TBAF (0.075 mL of a 1 M solution in THF, 0.075 mmol). The resulting solution was allowed to warm to room temperature (∼1 h) and diluted with CH_2Cl_2 (5 mL). The reaction mixture was poured into water (5 mL). After extraction with CH_2Cl_2 (2 × 5 mL), the organic layer was extracted with brine and dried over Na_2SO_4. Evaporation of the solvent followed by silica gel column chromatography of the resulting residue with a petroleum ether/EtOAc eluent afforded **6** in 99% yield (0.030 g) as a colorless oil (dr >99:1): R_f 0.6 (1:1 petroleum ether/EtOAc); $[\alpha]_D^{24}$ −90.25 (*c* 1.0, $CHCl_3$); IR (neat) ν_{max} 3260, 2924, 1731, 1466, 1063 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.48 (d, *J* = 8.0 Hz, 2H), 7.28−7.26 (m, 1H), 7.11 (d, *J* = 7.9 Hz, 3H), 6.92 (t, *J* = 7.4 Hz, 1H), 6.57 (d, *J* = 8.3 Hz, 1H), 5.22 (d, *J* = 3.8 Hz, 1H), 4.62 (s, 1H), 3.28 (s, 1H), 2.36 (s, 3H), 1.84 (s, 3H), 1.56 (s, 3H), 1.13 (s, 9H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 152.4, 145.0, 134.1, 132.0, 130.3, 129.9, 129.3, 129.25, 122.0, 121.2, 118.6, 116.4, 65.1, 56.4, 49.9, 22.6, 21.7, 18.6, 17.6; HRMS (ESI) *m*/*z* [M + Na]$^+$ calcd for $C_{23}H_{29}NO_4S_2Na$ 470.1430, found 470.1434.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c01982.

Copies of the ^{1}H and ^{13}C NMR spectra of the new compounds (PDF)

Accession Codes

CCDC 2350080 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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■ DEDICATION

Dedicated to the late Prof. Dieter Hoppe, University of Muenster, Germany, a pioneer in lithiation and carbanion chemistry.

■ REFERENCES

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